Thursday Warm-up: Ohm's Law, Electromotive Force (emf), Motional emf

Prof. Jordan C. Hanson

November 20, 2025

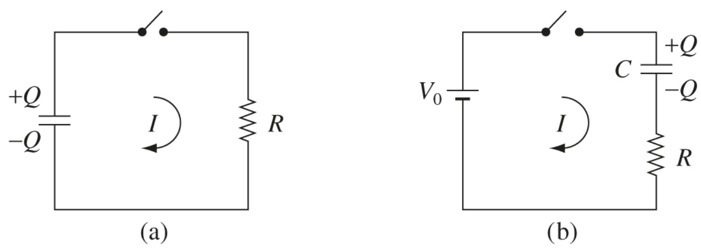

Figure 1: (Left) A charged capacitor drives current through a resistor. (Right) An emf charges a capacitor, and drives current through a resistor.

1 Memory Bank

- **Ohm's Law**, low-speed charges, small **B**-fields:

$$\mathbf{J} = \sigma \mathbf{f} = \sigma \mathbf{E} \tag{1}$$

The vector **f** is the force per unit charge, and σ is the *conductivity*. The parameter $1/\sigma = \rho$ is the *resistivity*.

- **Energy stored in a capacitor**:

$$U_{\mathrm{C}} = \frac{1}{2}CV^2 = \frac{1}{2}\frac{Q^2}{C} \tag{2}$$

- **Flux rule and emf**:

$$\mathcal{E} = -\frac{d\Phi}{dt} \tag{3}$$

$$\Phi = \int \mathbf{B} \cdot d\mathbf{a} \tag{4}$$

2 Ohm's Law and Motional emf

1. Starting with Eq. 1, show that a conductor with fixed cross-sectional area A, potential difference ΔV between the ends, and current I, that

$$\Delta V = IR \tag{5}$$

The parameter R should be related to the resistivity.

2. A capacitor C has been charged up to a potential V_0 (Fig. 1, left). At time $t = 0$, it is connected to a resistance R. (a) Determine the charge on the capacitor and the current as a function of time. (b) What is the original energy stored in the capacitor? (c) The power from a current I is $P = I^2R$. Integrate this to show that energy is conserved.

3. Determine $Q(t)$ and $I(t)$ for the emf in Fig. 1, right.

4. Suppose the force per unit charge is separated into the force per unit charge from the *source* (battery), and the electrostatic field smoothing the current:

$$\mathbf{f} = \mathbf{f}_s + \mathbf{E} \tag{6}$$

The work done on charges all the way around a circuit (emf) is $\oint \mathbf{f} \cdot d\mathbf{l}$. (a) Show that

$$\mathcal{E} = \oint \mathbf{f} \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l} \tag{7}$$

(b) Show that if the *net force* per unit charge **f** is zero, that $\Delta V = \mathcal{E}$.

5. Suppose the **B**-field inside a solenoid varies like $B(t) = \mu_0 n I(t)$, where $I(t) = at + b$. What is the emf induced on a concentric loop of wire with radius r inside the solenoid?